Summary of O-RAN's AI/ML Workflow Perspective

Architecture, Interfaces, Deployment Scenarios & Research Gaps

2026-06-23

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References: Polese, Bonati, D'Oro, Basagni, Melodia, *"Understanding O-RAN: Architecture, Interfaces, Algorithms, Security, and Research Challenges"* (arXiv:2202.01032, 2022). This document [reports the content of the paper with **AI/ML workflow as the axis** rearrange and the **research gap** derived therefrom Organize together.

1 Introduction: Why the "AI/ML Workflow" Axis?

The essence of O-RAN lies in disaggregating base stations and opening them up to standard interfaces, but the **purpose** of its existence is to enable the deployment of ML to RAN control. Therefore, if O-RAN is viewed not as "disaggregating boxes" but as a matter of **"how to place an ML pipeline on which node of the network"**, the architecture, interfaces, and deployment scenarios align into a single consistent picture.

From this perspective, the core questions can be condensed into two.

1. What **functional stages** make up an AI/ML workflow?
2. Each step **which node (SMO / Non-RT RIC / Near-RT RIC / RAN node)** can be placed at |||, and what is the constraint that determines the choice?

The decisive variable determining the answer to the second question is **timescale** , and this is the axis that runs through the entire document.

2 Background: O-RAN control layer hosting AI/ML

we organize the control layer that defines "where" the workflow is placed based on the time scale.

Table 1: Timescale layer of O-RAN control loop

Hierarchy	Timescale	Hosting Unit	Interface	Primary Role
SMO / Non-RT RIC	> 1 s	rApp	A1, O1, O2	Policy/Model Management, Training, Intent Management
Near-RT RIC	10 ms – 1 s	xApp	E2	Real-time RRM control, inference
RAN node (CU/DU)	< 10 ms	(embedded)	—	Real-time control such as scheduling and HARQ

If this hierarchical structure is formalized as a nested control loop, the outer loop is slower and global, while the inner loop is faster and local. When the control cycle is denoted as T_ℓ, approximately

$$T_{\text{Non-RT}} \;>\; 1\,\text{s} \;\gg\; T_{\text{Near-RT}} \in [10\,\text{ms},\, 1\,\text{s}] \;\gg\; T_{\text{RT}} \;<\; 10\,\text{ms}$$

holds. **<10 ms real-time loops are an area not standardized in the current specification (further study),** and this fact is directly related to the subsequent research gap.

3 Lifecycle Stages of AI/ML Workflow

The AI/ML lifecycle defined by O-RAN WG2 is a cyclic pipeline from data to inference, consisting of the following stages.

1. **Data Preparation** — Collection · Preprocessing · Feature Engineering
2. **AI/ML Training** — Model Training (Offline is generally required)
3. **Validation & Deployment** — Deploy to target nodes after validation
4. **Inference** — Perform real-time inference with a trained model
5. **Continuous Operations** — Post-deployment performance monitoring and retraining triggers

The core operating principles are as follows.

- **Offline training required**: Discontinue the discontinuation of unverified models. Online learning is permitted only for fine-tuning purposes.
- **Current Limit of Inference Host**: Inference is limited to **RIC** in the current specification, and Inference in CU/DU is in a pending (further study) state.

4 Node Deployment of Workflow Function: Deployment Scenario (Fig. 14)

Fig. 14 of the paper classifies the above workflow into five scenarios based on which node the five function columns (Model Management / Continuous Operations / Data Preparation / Training / Inference) are placed . Since this is the key table of this document, it is summarized in detail.

4.1 Function placement by scenario

Table 2: O-RAN AI/ML Deployment Scenario (Fig. 14 reconfigured)

Scenario	Model Mgmt	Continuous Ops	Data Prep	Training	Inference
1.1	Non-RT	Non-RT	Non-RT	Non-RT	Non-RT
1.2	SMO	Non-RT	Non-RT / Near-RT	Non-RT	**Near-RT**
1.3	SMO	Non-RT	SMO	SMO	Non-RT
1.4	Non-RT	**Near-RT**	Non-RT	Non-RT / SMO	**Near-RT**
1.5 (FFS)	Non-RT	(Undetermined)	(Undetermined)	(Undetermined)	**O-CU / O-DU**

4.2 Scenario Interpretation

- **1.1 — Fully Centralized**: All functions concentrated in a Non-RT RIC. Inference >Since it is on a 1 s scale, it is suitable for slow decisions such as slicing policies and traffic forecasting. Fast closed-loop control impossible.
- **1.2 — Separation of Training/Inference**: Training is Non-RT (Offline), **Inference is Near-RT** descends to form a 10 ms–1 s closed loop. The most commonly discussed structure.
- **1.3 — Separation of Authority**: Model Management and Training are owned by **SMO**, Operation and inference are handled by Non-RT. A structure in which a central SMO deploys a common model in a large-scale network and local Non-RTs perform inference.
- **1.4 — Online Adaptive Loop**: Unlike 1.2, up to **Continuous Operations** Near-RT** is in charge. The model self-monitors in real-time and requests retraining from Non-RT if performance degrades. It has the highest responsiveness among RIC-based scenarios.
- **1.5 (FFS) — RAN Node Inference**: **Inference is executed on the O-CU/O-DU itself**. The only structure that enables real-time control of the TTI scale, but is **not yet standardized (For Further Study)**.

  How to Read: In Scenario 1.5, which of the SMO · Near-RT RICs is empty in the table It is not "not needed", but rather **since it is FFS, the placement of that function is simply undefined**. Two controllers are always present in the network for FCAPS management, RRM, etc.

5 Chained AI/ML Models and Data Access Platform

emphasizes that the workflow can be configured using **model chaining** rather than a single model . The inference output of one model becomes the input of another model, and a data access platform (middleware) mediates the production/consumption of this.

Representative example (based on papers):

$$\underbrace{\text{rApp:}\quad\quad}_{A} \rightarrow \underbrace{\text{xApp Y:}\quad\quad}_{C} \rightarrow \underbrace{\text{xApp Z:}\quad\quad}_{D=f(A,B_{\text{KPM}},C)}$$

This structure illustrates a hierarchical information flow where **slow prediction (upper) sets the input for fast scheduling (lower)**, leading to the subsequent discussion on hierarchical design.

6 Use Cases Where ML Is Actually Applied

Among the ML application areas classified by the paper, the key points from the perspective of resource control are as follows.

- **Resource Allocation** (Network Slicing/Scheduling) — ML Application Most active area. Numerous closed-loop slicing control examples (ColO-RAN, NexRAN, etc.).
- **Mobility** — Handover/Load Balancing Prediction.
- **QoE/QoS Control** — Resource adjustment based on user perceived quality.
- **RAN Sharing / Massive MIMO / New Applications**(UAV · Vehicle · Industrial IoT).

Among these, **scheduling** is the region with the fastest time scale, so it is highlighted that only **1.5 (in-node inference)** is timing-appropriate in the scenario table above.

7 Research Gap

We address the gaps explicitly and implicitly left by the paper from the perspective of AI/ML workflow. Everything converges around **Scenario 1.5 (In-node Inference)**.

7.1 Gap 1 — Real-time (<10 ms) Absence of inference loop

Since the inference host is limited to RIC (10 ms), there is no standard path to directly apply ML to **fast control at the TTI scale** (scheduling, link adaptation). Formally, for in-node inference to be possible, the sum of the inference delay τ_{inf} and the scheduling processing delay τ_{sched} must not exceed the TTI budget .

$$\tau_{\text{inf}} + \tau_{\text{sched}} \;\leq\; \text{TTI} \quad (\text{TTI} \approx 0.5\text{–}1\,\text{ms})$$

It is not verified whether this inequality holds in the current hardware and model structure, and this is the first technical reason why 1.5 is FFS.

7.2 Gap 2 — Absence of model delivery interface

the 1.5 architecture where "Model Management is Non-RT and Inference is O-DU," a **standard interface to directly drop trained model weights from Non-RT to the node** is required. However, the current interface serves a different purpose.

Table 3: Standard space in model deployment path

Interface	Design Purpose	Model Deployment Suitability
A1	Policy · enrichment delivery	Partial
E2	RRM Control Command	Non-conforming (Not for Model File Transfer)
O1	FCAPS Management	Non-conformity

7.3 Gap 3 — Resource isolation

In-node inference implies that RAN functions and ML workloads **share the same hardware** . Consequently, (i) contention for GPU/CPU resources, (ii) jitter that undermines RAN real-time performance, and (iii) the absence of QoS-guaranteed resource allocation policies remain issues . A structure for **dynamic separation and allocation of RAN workloads and AI inference workloads** has not been defined.

7.4 Gap 4 — Lack of utilization of cross-layer signals

Cross-layer mapping that propagates signals from model layers (e.g., token importance of generative models — entropy, logprob, attention) as **utility inputs to MAC schedulers** is an area rarely covered in either standards or literature. This gains significance when combined with Gap 1 (fast control + model-derived signals).

8 Summary: Alignment to research positioning

The gaps above converge on a single axis called the **unrealized scenario 1.5**, and research demonstrating this naturally organizes into **hierarchical control** .

```
Non-RT RIC   →         /    prior       (> 1 s)
       A1
Near-RT RIC  →    ·    /             (10 ms-1 s)
       E2
O-DU (cuMAC) →    +       PRB             (< 10 ms)
```

- **The Slow Layer (RIC)** supplies policies, weights, and predictions to the Fast Layer.
- **Fast layer (node)** directly executes inference + scheduling at the TTI scale.

In this structure, the systematic contribution of the study lies in the third row — the inference + scheduling loop that actually runs at the node — which corresponds to the first demonstration of 1.5 remaining as FFS. The necessary technical elements (channel quality prediction, traffic prediction) are each mapped as inputs to the Near-RT layer, and the verification environment (GH200 / pyAerial / cuMAC / ROS2) is mapped as the actual answer for Gap $1 \cdot 3$.

Table 4: Mapping of Research Gap and Research Factors

Research Gap	Corresponding Factors	Verification Methods
Gap 1 (real-time inference)	sub-ms inference feasibility	GH200 + pyAerial measured
Gap 3 (Resource Isolation)	RAN+AI Workload Coexistence	Aerial Dynamic Allocation
Gap 4 (cross-layer signal)	Token Importance → Utility Input	cuMAC Scheduler
Prediction Factors	Channel/Traffic Prediction	Near-RT Layer Model

9 One-line summary

The O-RAN AI/ML workflow is a problem of deploying 5 functions to 4 nodes, and The variable that determines that choice is the time scale. **Descending inference to the RAN node Only Scenario 1.5 enables TTI scale control, but it still remains as FFS. exists**, and precisely this unrealized point is cross-layer · systems-level research It is the core void to be placed.